

Non-Functional Requirements

Requirement Analysis | Import Data Using Transform Maps

1. Introduction

Non-functional requirements describe how well the import solution must perform. They apply to every import run and are verified during testing.

2. Non-Functional Requirements

ID	Category	Requirement	Measure
NFR-01	Accuracy	Imported values match the source data	100% of valid rows match source
NFR-02	Data integrity	No duplicate records after repeated loads	0 duplicates on re-load
NFR-03	Performance	Files of several thousand rows load without timeout	Up to 10,000 rows per run
NFR-04	Security	Only authorised roles may import and transform	admin / import_transformer roles only
NFR-05	Auditability	Every run is logged	Import set and transform history kept
NFR-06	Reliability	A failed row does not stop the full import	Errors logged, other rows continue
NFR-07	Usability	Setup is simple with clear steps	New admin completes import with the guide
NFR-08	Maintainability	Scripts are short and commented	Each script has comments and a purpose
NFR-09	Reusability	Same map works for future files	Layout change requires only map edits
NFR-10	Compatibility	Supports CSV, XLS and XLSX	All three formats load correctly
NFR-11	Data privacy	Personal data handled as per policy	Files removed after import where required
NFR-12	Recoverability	Bad import can be corrected	Backup taken; records identified by import set

3. Quality Attribute Priorities

Priority	Attribute	Reason
1	Accuracy and integrity	Bad data affects reports and workflows
2	Security	Imports can change many records at once
3	Auditability	Needed for compliance and troubleshooting
4	Performance	Matters for larger files
5	Usability and reuse	Reduces effort over time

4. Constraints Affecting These Requirements

- Work is done in a ServiceNow Personal Developer Instance (PDI) with limited resources.
- Only standard platform features are used, with no extra plugins or licences.
- Large files may need to be split to avoid timeouts.